Benjamin Poole

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A Google scholarship recipient and researcher studying MSc in Artificial Intelligence. Graduated in Computer Science (1st Class) and was offered a funded PhD from my research experience in agents and reinforcement learning.

Education

University of Edinburgh — Edinburgh, UK
MSc Artificial Intelligence — *September 2026 – Present*

University of Southampton — Southampton, UK
BEng Computer Science (1st Class) — *September 2023 – July 2026*

Research Experience

Researcher — March – September 2024
Agents Interaction and Complexity research group - University of Southampton
Researched the development of a vision language policy to control robot agents, supervised by a Postgraduate researcher.

- Prototyped an agent policy based on the MineCLIP dataset (arxiv:2206.08853) to build a teacher model that could train a student model using Imitation Learning.
- Invited back to research swarm robotics for XPRIZE Wildfires

Researcher — October 2022 – August 2023
Tupel 3D
Researched and prototyped a low-cost, structured light hand-held 3D scanner.

- Developed a proprietary AI to create a 3D depth reconstruction system that could provide accurate 3D imagery using only primitive hardware. The company believe that this system is unique and superior to all publications they know.
- Developed a unique SLAM camera tracking system following ORB-SLAM 3 (arxiv:2007.11898) that would allow fast camera tracking for 3D reconstruction.

Consulting

Software Developer — August – September 2025
Britwind Ecotricity

- Developed an IoT energy dashboard using web technologies, to display customer wind turbine power generation allowing full analytics and customisation in 6 weeks. This saved significant development costs by replacing a previously stalled, contractor-dependent effort.

Featured Projects

Multi-agent reinforcement learning for household robots (2026)

- Developed a multi-agent system using experiential reinforcement learning to train frozen agents in the long-horizon behaviour-1K environment, learning natural-language heuristics from experience, achieving state-of-the-art 16% success at roughly 1/1000th the training cost of prior work.

From-scratch implementation and analysis of an LLM (2024)

- Developed a 14.5m parameter language model from scratch (matrix multiplication), following the TinyStories paper (arXiv:2305.07759). The model was able to coherently model story language despite its small size.
- Analysed the model word vectors using 2D T-SNE dimensionality reduction to understand how the model interprets word semantics in terms of their context and relation to other words.

Sketch to image Diffusion model (2023)

- An implementation of the paper SDEdit (arxiv:2108.01073) converting MS Paint sketches to photorealistic images without model retraining.

Ferranti Mark 1 Assembler and Emulator (2025)

- Browser-based emulator of the 1951 Ferranti Mark 1, with an assembler from a small assembly language to the machine's native code, including its five-bit teleprinter encoding, with live visualisation of the Williams-tube electronic store.